(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(CBCS)
B. Tech CSE AI, Semester: 5
(Oct, 2022)

Subject Code: BAI-305 Subject: Deep Learning
Time : 1 ½ Hours Maximum Marks : 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 (4+3+3)

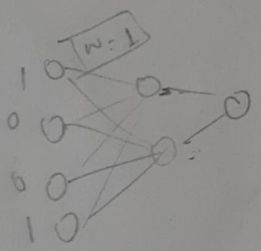
(a) Using McCulloch Pitts units, implement AND and OR gates. Draw neat diagrams.

(b) What are the two types of feature scaling techniques (explain with formula) for numeric data? Perform them on following 1-D data.
1, 3, 5, 1, 2, 3, 7, 1, 8

(c) Draw computational graph for equation: $J(\Theta) = 2\ (ab+c)$. Find all partial derivatives, and solve for values a=2, b=3, c=1

Q2 (5,5)

(a) Assume a two layer neural network. Number of inputs are four. Number of neurons in the first hidden layer are two (with ReLU as activation function), and one (with sigmoid as activation function) neuron in the second hidden layer. Assume all weights are 1. Inputs are 1, 0, 0, and 1. What is the output? Draw a neat diagram and compute output at each neuron.



(b) Using a diagram, explain the calculations (give equations) inside a neural network. Write steps for a perceptron learning algorithm.

Q3 (5,5)

(a) Suppose that the predicted output for 3-class classifiers is 4.5, 2.9, 1.2 and actual output is 0,1,0. Calculate cross entropy loss, MAE, and MSE.

(b) We usually start with a simple model, and then check for various conditions during training, explaining the entire process using a flow diagram.

Q4 (5,5)

(a) Using learning curves, explain the situations for over fitting and under fitting. How do you recover from each of them?

(b) Suppose output of a 4-class classifier is 3,2,-1,4. Compute softmax operation on it. How do you perform data augmentation for numeric, image, and text datasets?